

END-TO-END PHYSICAL AI ONCOLOGY CLINICAL TRIAL
UNIFICATION

Adaption: 21 CFR Part 50

Protection of Human Subjects

Modified ECFR

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Contents

1	SUBPART A – GENERAL PROVISIONS	1
1.1	§ 50.1 Scope	1
1.1.1	Physical AI Adaptation of § 50.1	1
1.2	§ 50.3 Definitions	2
1.2.1	Physical AI Definitions	5
2	SUBPART B – INFORMED CONSENT OF HUMAN SUBJECTS	7
2.1	§ 50.20 General Requirements for Informed Consent	7
2.1.1	Physical AI Adaptation of § 50.20	8
2.2	§ 50.22 Exception from Informed Consent Requirements for Minimal Risk Clinical Investigations	9
2.2.1	Physical AI Adaptation of § 50.22	9
2.3	§ 50.23 Exception from General Requirements	10
2.3.1	Physical AI Adaptation of § 50.23	10
2.3.2	Prior 21 CFR Part 50: § 50.23 Military Exception Provisions	11
2.4	§ 50.24 Exception from Informed Consent Requirements for Emergency Research . .	11
2.4.1	Physical AI Adaptation of § 50.24	13
2.5	§ 50.25 Elements of Informed Consent	14
2.5.1	Physical AI Adaptation of § 50.25 Basic Elements	15
2.5.2	Physical AI Additional Elements of Informed Consent	16
2.6	§ 50.27 Documentation of Informed Consent	18
2.6.1	Physical AI Adaptation of § 50.27	18
3	SUBPART C – ADDITIONAL PROTECTIONS FOR SUBJECTS IN PHYSICAL AI CLINICAL INVESTIGATIONS	19
3.1	§ 50.30 Physical AI System Safety Requirements	19
3.1.1	Pre-Procedure Safety Matrix	19
3.1.2	Runtime Safety Monitoring	20
3.1.3	Post-Procedure Requirements	21
3.1.4	Task-Order Lifecycle	21
3.1.5	Forbidden Operations	22
3.2	§ 50.31 IRB Review of Physical AI Investigations	22
3.3	§ 50.32 Ongoing Consent and Subject Notification	23
3.4	§ 50.33 Data Protection for Physical AI Investigations	24
3.5	§ 50.34 Physical AI System Classification and Regulatory Pathways	25
4	SUBPART D – ADDITIONAL SAFEGUARDS FOR CHILDREN IN CLINICAL INVESTIGATIONS	26
4.1	§ 50.50 IRB Duties	26
4.1.1	Physical AI Adaptation of § 50.50	26
4.2	§ 50.51 Clinical Investigations Not Involving Greater Than Minimal Risk	26
4.2.1	Physical AI Adaptation of § 50.51	26
4.3	§ 50.52 Clinical Investigations Involving Greater Than Minimal Risk but Presenting the Prospect of Direct Benefit	26

4.3.1	Physical AI Adaptation of § 50.52	27
4.4	§ 50.53 Clinical Investigations Involving Greater Than Minimal Risk and No Prospect of Direct Benefit	27
4.4.1	Physical AI Adaptation of § 50.53	27
4.5	§ 50.54 Clinical Investigations Not Otherwise Approvable	27
4.5.1	Physical AI Adaptation of § 50.54	28
4.6	§ 50.55 Requirements for Permission by Parents or Guardians and for Assent by Children	28
4.6.1	Physical AI Adaptation of § 50.55	29
4.7	§ 50.56 Wards	29
4.7.1	Physical AI Adaptation of § 50.56	29
5	GLOSSARY	30

Prefatory Note

This adaptation extends the prior 21 CFR Part 50 – Protection of Human Subjects to address the end-to-end integration of Physical AI systems, advanced robotics, digital twins, and AI/ML agents into oncology clinical trials. The prior 21 CFR Part 50 regulation established foundational protections for human subjects in clinical investigations regulated by the Food and Drug Administration. This current document adapts those protections to encompass the specialized requirements of Physical AI oncology clinical trial unification, including autonomous surgical robots, therapeutic positioning systems, diagnostic needle-placement platforms, rehabilitative exoskeletons, companion monitoring systems, AI-driven decision support, and simulation-validated clinical procedures.

The Unification Standard Level (USL) scoring framework (v1.4.0 through v1.8.0) provides quantitative readiness assessments for robotic platforms across four dimensions: simulation switching, AI integration, cross-robot sharing, and clinical trial collaboration. USL scores range from 1.0 to 10.0 and are referenced throughout this adaptation to establish minimum readiness thresholds for clinical deployment and to inform risk-based consent requirements.

The national MCP-PAI standard (v1.2.0) defines the five-server Model Context Protocol topology (authorization, FHIR, DICOM, ledger, provenance), 23 standardized tools, six actor roles, and five cumulative conformance levels that govern data access, audit, and robot agent operations in Physical AI oncology trials. This protocol infrastructure underpins the data protection and audit trail requirements throughout this adaptation.

This adaptation should be read in conjunction with the physical-ai-oncology-trials repository (v2.3.0, DOI: 10.5281/zenodo.18445179), the national MCP-PAI oncology trials standard (v1.2.0, DOI: 10.5281/zenodo.18869776), the ICH E6(R3) Physical AI adaptation (DOI: 10.5281/zenodo.18973368), and the federated learning framework (DOI: 10.5281/zenodo.18840880).

Document History

Prior Regulation 21 CFR Part 50 – Protection of Human Subjects. Originally published at 45 FR 36390, May 30, 1980, with subsequent amendments through 2023. The prior regulation established the baseline framework for protecting human subjects in FDA-regulated clinical investigations.

Current Adaptation End-to-End Physical AI Oncology Clinical Trial Unification: Adaption of 21 CFR Part 50, released 16 March 2026. This adaptation incorporates Physical AI requirements throughout for oncology clinical investigations involving autonomous robotic systems, digital twins, and AI/ML agents.

Repository Refs physical-ai-oncology-trials v2.3.0 (March 2026), national-mcp-pai-oncology-trials v1.2.0 (March 2026). These repositories provide validated code, tools, specifications, and documentation supporting this adaptation.

Prior 21 CFR Part 50: Public Domain Notice

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Authority: 21 U.S.C. 321, 343, 346, 346a, 348, 350a, 350b, 352, 353, 355, 360, 360c–360f, 360h–360j, 371, 379e, 381; 42 U.S.C. 216, 241, 262.

Source: 45 FR 36390, May 30, 1980, unless otherwise noted.

1 SUBPART A – GENERAL PROVISIONS

1.1 § 50.1 Scope

(a) This part applies to all clinical investigations regulated by the Food and Drug Administration under sections 505(i) and 520(g) of the Federal Food, Drug, and Cosmetic Act, as well as clinical investigations that support applications for research or marketing permits for products regulated by the Food and Drug Administration, including foods, including dietary supplements, that bear a nutrient content claim or a health claim, infant formulas, food and color additives, drugs for human use, medical devices for human use, biological products for human use, and electronic products. Additional specific obligations and commitments of, and standards of conduct for, persons who sponsor or monitor clinical investigations involving particular test articles may also be found in other parts (e.g., parts 312 and 812). Compliance with these parts is intended to protect the rights and safety of subjects involved in investigations filed with the Food and Drug Administration pursuant to sections 403, 406, 409, 412, 413, 502, 503, 505, 510, 513–516, 518–520, 721, and 801 of the Federal Food, Drug, and Cosmetic Act and sections 351 and 354–360F of the Public Health Service Act.

(b) References in this part to regulatory sections of the Code of Federal Regulations are to chapter I of title 21, unless otherwise noted.

1.1.1 Physical AI Adaptation of § 50.1

(c) This part additionally applies to all clinical investigations in which Physical AI systems perform, assist, or monitor clinical procedures on human subjects. Physical AI systems encompass the five robot types defined in the national MCP-PAI standard (v1.2.0): autonomous surgical robots, therapeutic positioning systems, diagnostic needle-placement platforms, rehabilitative exoskeletons, and companion monitoring systems. These systems operate through MCP-mediated data access with deny-by-default authorization, hash-chained audit trails, and HIPAA Safe Harbor de-identification protections.

(d) The scope of protection under this part extends to subjects whose clinical data is processed by Physical AI systems, including data used for digital twin modeling, AI/ML inference, federated learning, and simulation-based procedure planning. The processing of subject data through the five MCP servers (trialmcp-authz, trialmcp-fhir, trialmcp-dicom, trialmcp-ledger, trialmcp-provenance) constitutes an interaction subject to the consent and protection requirements of this part.

(e) The Unification Standard Level (USL) framework serves as the readiness assessment methodology for Physical AI platforms used in clinical investigations. USL scores are computed across four equally weighted dimensions (simulation switching, AI integration, cross-robot sharing, clinical trial collaboration) and range from 1.0 (Minimal) to 10.0 (Reference). The national MCP-PAI standard specifies minimum USL thresholds by procedure type: surgical robots require USL 7.0 or above, therapeutic positioning and diagnostic needle placement require USL 6.0 or above, rehabilitative exoskeletons require USL 4.0 or above, and companion monitoring systems require USL 3.0 or above.

(f) The seven robot categories recognized for clinical investigation purposes, consistent with the physical-ai-oncology-trials repository (v2.3.0) and USL scoring methodology, are: surgical robots, cobots, humanoid robots, therapeutic robots, diagnostic robots, assistive robots, and rehabilitative robots. Nine robotic platforms have been evaluated under the USL framework: da Vinci dVRK (USL 7.1), Franka Emika Panda (USL 7.4), Boston Dynamics Atlas (USL 5.8), Kinova Gen3 (USL 5.7), Medtronic Hugo RAS (USL 4.5), Agility Digit (USL 4.2), Tesla Optimus (USL 3.6), CMR Versius (USL 3.4), and UFACTORY xArm 7 (USL 3.4).

(g) Five categories of AI/ML are recognized for use in Physical AI clinical investigations: generative AI (vision-language-action models, diffusion policies, synthetic data generation), agentic AI (LLM-based assistants, multi-agent coordination via CrewAI v1.6.1 and LangGraph v1.1.0), reinforcement learning (sim-to-real transfer, hierarchical and multi-agent RL via NVIDIA Isaac Lab v2.3.1), self-supervised learning (contrastive learning, foundation model pretraining), and supervised learning (segmentation, detection, classification via MONAI).

(h) Four simulation frameworks are recognized for pre-clinical validation of Physical AI systems: NVIDIA Isaac Lab (v2.3.1) for GPU-accelerated robot training, MuJoCo (v3.4.0) for precision physics simulation, Gazebo Sim (v10.0.0) for ROS 2 integrated simulation, and PyBullet (v3.2.5) for rapid prototyping. Cross-framework validation via the unification bridge must demonstrate behavioral consistency with trajectory deviations below 2mm and force discrepancies below 0.5N.

[45 FR 36390, May 30, 1980; 46 FR 8979, Jan. 27, 1981, as amended at 63 FR 26697, May 13, 1998; 64 FR 399, Jan. 5, 1999; 66 FR 20597, Apr. 24, 2001. Physical AI adaptation added 16 March 2026.]

1.2 § 50.3 Definitions

As used in this part:

(a) *Act* means the Federal Food, Drug, and Cosmetic Act, as amended (secs. 201–902, 52 Stat. 1040 *et seq.* as amended (21 U.S.C. 321–392)).

(b) *Application for research or marketing permit* includes:

- (1) A color additive petition, described in part 71.
- (2) A food additive petition, described in parts 171 and 571.
- (3) Data and information about a substance submitted as part of the procedures for establishing that the substance is generally recognized as safe for use that results or may reasonably be expected to result, directly or indirectly, in its becoming a component or otherwise affecting the characteristics of any food, described in §§ 170.30 and 570.30.

- (4) Data and information about a food additive submitted as part of the procedures for food additives permitted to be used on an interim basis pending additional study, described in § 180.1.
- (5) Data and information about a substance submitted as part of the procedures for establishing a tolerance for unavoidable contaminants in food and food-packaging materials, described in section 406 of the act.
- (6) An investigational new drug application, described in part 312 of this chapter.
- (7) A new drug application, described in part 314.
- (8) Data and information about the bioavailability or bioequivalence of drugs for human use submitted as part of the procedures for issuing, amending, or repealing a bioequivalence requirement, described in part 320.
- (9) Data and information about an over-the-counter drug for human use submitted as part of the procedures for classifying these drugs as generally recognized as safe and effective and not misbranded, described in part 330.
- (10) Data and information about a prescription drug for human use submitted as part of the procedures for classifying these drugs as generally recognized as safe and effective and not misbranded, described in this chapter.
- (11) [Reserved]
- (12) An application for a biologics license, described in part 601 of this chapter.
- (13) Data and information about a biological product submitted as part of the procedures for determining that licensed biological products are safe and effective and not misbranded, described in part 601.
- (14) Data and information about an in vitro diagnostic product submitted as part of the procedures for establishing, amending, or repealing a standard for these products, described in part 809.
- (15) *An Application for an Investigational Device Exemption*, described in part 812.
- (16) Data and information about a medical device submitted as part of the procedures for classifying these devices, described in section 513.
- (17) Data and information about a medical device submitted as part of the procedures for establishing, amending, or repealing a standard for these devices, described in section 514.
- (18) An application for premarket approval of a medical device, described in section 515.
- (19) A product development protocol for a medical device, described in section 515.
- (20) Data and information about an electronic product submitted as part of the procedures for establishing, amending, or repealing a standard for these products, described in section 358 of the Public Health Service Act.
- (21) Data and information about an electronic product submitted as part of the procedures for obtaining a variance from any electronic product performance standard, as described in § 1010.4.

- (22) Data and information about an electronic product submitted as part of the procedures for granting, amending, or extending an exemption from a radiation safety performance standard, as described in § 1010.5.
- (23) Data and information about a clinical study of an infant formula when submitted as part of an infant formula notification under section 412(c) of the Federal Food, Drug, and Cosmetic Act.
- (24) Data and information submitted in a petition for a nutrient content claim, described in § 101.69 of this chapter, or for a health claim, described in § 101.70 of this chapter.
- (25) Data and information from investigations involving children submitted in a new dietary ingredient notification, described in § 190.6 of this chapter.

(c) *Clinical investigation* means any experiment that involves a test article and one or more human subjects and that either is subject to requirements for prior submission to the Food and Drug Administration under section 505(i) or 520(g) of the act, or is not subject to requirements for prior submission to the Food and Drug Administration under these sections of the act, but the results of which are intended to be submitted later to, or held for inspection by, the Food and Drug Administration as part of an application for a research or marketing permit. The term does not include experiments that are subject to the provisions of part 58 of this chapter, regarding nonclinical laboratory studies.

(d) *Investigator* means an individual who actually conducts a clinical investigation, i.e., under whose immediate direction the test article is administered or dispensed to, or used involving, a subject, or, in the event of an investigation conducted by a team of individuals, is the responsible leader of that team.

(e) *Sponsor* means a person who initiates a clinical investigation, but who does not actually conduct the investigation, i.e., the test article is administered or dispensed to or used involving, a subject under the immediate direction of another individual. A person other than an individual (e.g., corporation or agency) that uses one or more of its own employees to conduct a clinical investigation it has initiated is considered to be a sponsor (not a sponsor-investigator), and the employees are considered to be investigators.

(f) *Sponsor-investigator* means an individual who both initiates and actually conducts, alone or with others, a clinical investigation, i.e., under whose immediate direction the test article is administered or dispensed to, or used involving, a subject. The term does not include any person other than an individual, e.g., corporation or agency.

(g) *Human subject* means an individual who is or becomes a participant in research, either as a recipient of the test article or as a control. A subject may be either a healthy human or a patient.

(h) *Institution* means any public or private entity or agency (including Federal, State, and other agencies). The word *facility* as used in section 520(g) of the act is deemed to be synonymous with the term *institution* for purposes of this part.

(i) *Institutional review board* (IRB) means any board, committee, or other group formally designated by an institution to review biomedical research involving humans as subjects, to approve the initiation of and conduct periodic review of such research. The term has the same meaning as the phrase *institutional review committee* as used in section 520(g) of the act.

(j) *Test article* means any drug (including a biological product for human use), medical device for human use, human food additive, color additive, electronic product, or any other article subject to regulation under the act or under sections 351 and 354–360F of the Public Health Service Act (42 U.S.C. 262 and 263b–263n).

(k) *Minimal risk* means that the probability and magnitude of harm or discomfort anticipated in the research are not greater in and of themselves than those ordinarily encountered in daily life or during the performance of routine physical or psychological examinations or tests.

(l) *Legally authorized representative* means an individual or judicial or other body authorized under applicable law to consent on behalf of a prospective subject to the subject’s participation in the procedure(s) involved in the research.

(m) *Family member* means any one of the following legally competent persons: Spouse; parents; children (including adopted children); brothers, sisters, and spouses of brothers and sisters; and any individual related by blood or affinity whose close association with the subject is the equivalent of a family relationship.

(n) *Assent* means a child’s affirmative agreement to participate in a clinical investigation. Mere failure to object should not, absent affirmative agreement, be construed as assent.

(o) *Children* means persons who have not attained the legal age for consent to treatments or procedures involved in clinical investigations, under the applicable law of the jurisdiction in which the clinical investigation will be conducted.

(p) *Parent* means a child’s biological or adoptive parent.

(q) *Ward* means a child who is placed in the legal custody of the State or other agency, institution, or entity, consistent with applicable Federal, State, or local law.

(r) *Permission* means the agreement of parent(s) or guardian to the participation of their child or ward in a clinical investigation.

(s) *Guardian* means an individual who is authorized under applicable State or local law to consent on behalf of a child to general medical care.

1.2.1 Physical AI Definitions

The following additional definitions apply to clinical investigations involving Physical AI systems. These definitions are derived from the physical-ai-oncology-trials repository (v2.3.0) and the national MCP-PAI oncology trials standard (v1.2.0).

(t) *Physical AI system* means an AI system operating in the physical world through a robotic platform used in clinical investigations. Physical AI systems include autonomous surgical robots, therapeutic positioning systems, diagnostic needle-placement platforms, rehabilitative exoskeletons, and companion monitoring systems. The term encompasses the robotic hardware, its control software, embedded AI/ML models, and associated sensor systems that collectively perform, assist, or monitor clinical procedures on human subjects.

(u) *Robot agent* means an autonomous Physical AI system executing clinical procedures within an oncology trial, interacting through MCP-mediated data access. The robot agent operates under the six-step workflow defined in the national MCP-PAI standard: authenticate, retrieve clinical data,

access imaging, execute procedure, record audit, and record provenance. Robot agents are one of six actor types (alongside Trial Coordinator, Data Monitor, Auditor, Sponsor, and CRO) in the MCP actor model.

(v) *Unification Standard Level (USL)* means a scoring framework (v1.4.0 through v1.8.0) evaluating Physical AI platform readiness across four equally weighted dimensions: simulation switching, AI integration, cross-robot sharing, and clinical trial collaboration. USL scores range from 1.0 to 10.0, mapping to ten levels from Minimal (1.0) to Reference (10.0) and three readiness bands: Foundational (1.0 to 4.9), Intermediate (5.0 to 6.9), and Advanced (7.0 to 10.0).

(w) *Digital twin* means a patient-specific computational model used for treatment simulation, procedure planning, and real-time intraoperative guidance. Digital twins are calibrated to longitudinal imaging data using reaction-diffusion equations, pharmacokinetic/pharmacodynamic models, or other mathematical frameworks and are continuously updated with real-world clinical data.

(x) *Simulation framework* means a physics-based computational environment used for validating Physical AI system behavior before clinical deployment. Recognized simulation frameworks include NVIDIA Isaac Lab (v2.3.1), MuJoCo (v3.4.0), Gazebo Sim (v10.0.0), and PyBullet (v3.2.5). The validated integration path follows: training (Isaac Lab) to validation (MuJoCo) to hardware deployment to clinical use.

(y) *Robot capability profile* means a machine-readable specification of a Physical AI platform's capabilities, safety prerequisites, USL score, and required MCP tools. The profile is defined by a JSON schema (robot-capability-profile.schema.json) in the national MCP-PAI standard and must be registered with the authorization server before any clinical procedure.

(z) *Task order* means a scheduled clinical trial task assigned to a Physical AI system, including procedure type, robot assignment, safety check requirements, and lifecycle state. Task orders progress through defined states: scheduled, safety_check, in_progress, and completed, cancelled, or failed.

(aa) *Emergency stop* means an immediate halt capability that must be available for any Physical AI system during clinical procedures, with mandatory audit recording. Activation of an emergency stop transitions the task order to a failed state and requires a new safety check cycle before any subsequent procedure.

(bb) *Human oversight* means the requirement that qualified clinical personnel maintain supervisory authority over Physical AI systems during all phases of clinical investigation. Human oversight encompasses the ability to monitor, intervene, override, or terminate Physical AI system operations at any time. The human oversight quality management framework (v1.2.0) specifies risk-tiered controls with escalating requirements based on clinical consequence.

(cc) *Hash-chained audit trail* means an immutable, tamper-evident record of all Physical AI system actions using SHA-256 hash chains. Each audit record contains the action performed, actor identity, timestamp, resource affected, and a hash linking it to the previous record. This mechanism satisfies the electronic records requirements of 21 CFR Part 11.

(dd) *Model Context Protocol (MCP)* means an open protocol for connecting AI agents to external tools and data sources, implemented through five standardized servers: trialmcp-authz (authorization), trialmcp-fhir (clinical data), trialmcp-dicom (imaging), trialmcp-ledger (audit), and trialmcp-

provenance (data lineage). The protocol defines 23 tools and operates under the governance of the Linux Foundation AI and Data Foundation.

(ee) *Deny-by-default authorization* means a security model where all Physical AI system access to clinical data is denied unless explicitly permitted by policy. Token-based sessions with defined scopes and time limits govern every data access request. This authorization model is enforced by the trialmcp-authz server.

(ff) *HIPAA Safe Harbor de-identification* means the removal or generalization of all 18 Protected Health Information (PHI) identifiers before clinical data is provided to Physical AI systems. The 18 identifiers include names, geographic data, dates, phone numbers, fax numbers, email addresses, Social Security numbers, medical record numbers, health plan beneficiary numbers, account numbers, certificate/license numbers, vehicle identifiers, device identifiers, web URLs, IP addresses, biometric identifiers, full-face photographs, and any other unique identifying number. HMAC-SHA256 with site-specific salts is used for pseudonymization.

(gg) *Predetermined Change Control Plan (PCCP)* means a documented plan for managing modifications to Physical AI systems used as or within medical devices, per FDA AI/ML guidance. The PCCP specifies the types of anticipated modifications, the methodology for implementing and validating changes, and the assessment of the impact on safety and effectiveness.

(hh) *Federated learning* means multi-site AI model training without sharing raw patient data, using differential privacy (epsilon and delta parameters) and secure aggregation (additive secret sharing, pairwise masking). The federation framework supports FedAvg, FedProx, and SCAFFOLD algorithms with data harmonization for DICOM, FHIR, ICD-10 to SNOMED CT, and LOINC tumor marker standardization.

(ii) *Pre-procedure safety matrix* means the set of verification checks that must pass before a Physical AI system may begin a clinical procedure. The matrix includes: valid authorization token, patient identity verification, clinical data availability (FHIR), imaging data access (DICOM), robot readiness and USL threshold verification, environmental checks, simulation validation confirmation, and digital twin synchronization status.

(jj) *Conformance level* means one of five cumulative levels defined in the national MCP-PAI standard for MCP server implementations: Core (base audit and authorization), Clinical Read (FHIR R4 clinical data access), Imaging (DICOM imaging with RECIST), Federated Site (multi-site federated learning with differential privacy), and Robot Procedure (full robot agent workflow with safety matrix and USL integration).

[45 FR 36390, May 30, 1980, as amended through 78 FR 12950, Feb. 26, 2013. Physical AI definitions added 16 March 2026.]

2 SUBPART B – INFORMED CONSENT OF HUMAN SUBJECTS

Source: 46 FR 8951, Jan. 27, 1981, unless otherwise noted.

2.1 § 50.20 General Requirements for Informed Consent

Except as provided in §§ 50.22, 50.23, and 50.24, no investigator may involve a human being as a subject in research covered by these regulations unless the investigator has obtained the legally

effective informed consent of the subject or the subject's legally authorized representative. An investigator shall seek such consent only under circumstances that provide the prospective subject or the representative sufficient opportunity to consider whether or not to participate and that minimize the possibility of coercion or undue influence. The information that is given to the subject or the representative shall be in language understandable to the subject or the representative. No informed consent, whether oral or written, may include any exculpatory language through which the subject or the representative is made to waive or appear to waive any of the subject's legal rights, or releases or appears to release the investigator, the sponsor, the institution, or its agents from liability for negligence.

2.1.1 Physical AI Adaptation of § 50.20

In addition to the general requirements above, informed consent for clinical investigations involving Physical AI systems must address the following:

- (a) The informed consent process must disclose the role of all autonomous robotic systems, AI/ML models, and digital twin technologies that will interact with or affect the subject during the investigation. Disclosure must include whether the Physical AI system will make autonomous clinical decisions, serve in an assistive capacity under human teleoperation, or operate in a shared-autonomy mode.
- (b) The degree of human oversight during procedures must be clearly explained, including the identity and qualifications of the clinical personnel maintaining supervisory authority, the mechanism by which the supervising clinician can intervene or override the Physical AI system, and the conditions under which an emergency stop would be activated.
- (c) Simulation validation results relevant to the subject's planned procedure must be made available upon request. The informed consent document must state whether the specific procedure has been validated in simulation, which simulation framework(s) were used, and what the cross-framework consistency results demonstrated.
- (d) The USL readiness score of the Physical AI platform assigned to the subject's procedure must be disclosed. The consent document must explain the USL score in plain language, including what the score represents, the minimum threshold required for the procedure type, and how the assigned platform compares to the threshold.
- (e) The specific robot type(s) to be used must be identified by manufacturer, model designation, and clinical function. The safety record of the platform must be summarized, including the number of prior clinical procedures, any reported safety incidents, and the emergency stop provisions.
- (f) If a digital twin model will be created from the subject's data, the consent document must describe what patient data will be used to construct the twin, how the twin will be used for treatment planning or procedure guidance, whether the twin model will be retained after the investigation, and the subject's right to request deletion of the twin model.
- (g) The consent document must explain how the subject's data will be processed through the five MCP servers, including the HIPAA Safe Harbor de-identification protections applied before data reaches Physical AI systems, the deny-by-default authorization model governing data access, and the hash-chained audit trail that records all data access events.

[46 FR 8951, Jan. 27, 1981, as amended at 64 FR 10942, Mar. 8, 1999; 88 FR 88248, Dec. 21, 2023.

Physical AI adaptation added 16 March 2026.]

2.2 § 50.22 Exception from Informed Consent Requirements for Minimal Risk Clinical Investigations

The IRB may approve a consent procedure that omits or alters some or all of the elements of informed consent set forth in § 50.25, or waive the requirements to obtain informed consent, provided that the IRB finds and documents that:

- (a) The clinical investigation involves no more than minimal risk to the subjects;
- (b) The clinical investigation could not practicably be carried out without the requested waiver or alteration;
- (c) If the clinical investigation involves using identifiable private information or identifiable biospecimens, the clinical investigation could not practicably be carried out without using such information or biospecimens in an identifiable format;
- (d) The waiver or alteration will not adversely affect the rights and welfare of the subjects; and
- (e) Whenever appropriate, the subjects or legally authorized representatives will be provided with additional pertinent information after participation.

2.2.1 Physical AI Adaptation of § 50.22

(f) When Physical AI systems are involved in a clinical investigation, the determination of minimal risk must consider the specific robot type and its operational characteristics. The following guidelines apply to the assessment of minimal risk in Physical AI investigations:

- (1) Companion monitoring systems (USL threshold 3.0 or above) that perform only non-contact observation, environmental monitoring, or verbal interaction with subjects may qualify for minimal risk classification, provided no physical contact with the subject occurs and no autonomous clinical decisions are made based on the monitoring data.
 - (2) Low-risk rehabilitative applications involving exoskeletons or gait trainers (USL threshold 4.0 or above) may qualify for minimal risk classification when the device operates within force limits that do not exceed those encountered in routine physical therapy, the subject can independently remove or deactivate the device at any time, and the device has been validated through at least 100 hours of simulation testing with documented safety performance.
 - (3) Surgical robots, therapeutic positioning systems, and diagnostic needle-placement platforms generally do not qualify for minimal risk classification due to the inherent physical contact, tissue interaction, or needle insertion involved in their operation. These systems require full informed consent as specified in §§ 50.20 and 50.25.
 - (4) Any Physical AI system that makes autonomous clinical decisions affecting treatment delivery, dose adjustment, or procedure modification does not qualify for minimal risk classification, regardless of its robot type or USL score.
- (g) The IRB must map the minimal risk determination to the USL score and conformance level of the Physical AI system. Systems operating at Conformance Level 5 (Robot Procedure) under the national MCP-PAI standard require full informed consent for any procedure involving physical

contact with the subject. Systems operating at Conformance Levels 1 through 4 may be evaluated for minimal risk on a case-by-case basis.

[88 FR 88248, Dec. 21, 2023. Physical AI adaptation added 16 March 2026.]

2.3 § 50.23 Exception from General Requirements

(a) The obtaining of informed consent shall be deemed feasible unless, before use of the test article (except as provided in paragraph (b) of this section), both the investigator and a physician who is not otherwise participating in the clinical investigation certify in writing all of the following:

- (1) The human subject is confronted by a life-threatening situation necessitating the use of the test article.
- (2) Informed consent cannot be obtained from the subject because of an inability to communicate with, or obtain legally effective consent from, the subject.
- (3) Time is not sufficient to obtain consent from the subject's legal representative.
- (4) There is available no alternative method of approved or generally recognized therapy that provides an equal or greater likelihood of saving the life of the subject.

(b) If immediate use of the test article is, in the investigator's opinion, required to preserve the life of the subject, and time is not sufficient to obtain the independent determination required in paragraph (a) of this section in advance of using the test article, the determinations of the clinical investigator shall be made and, within 5 working days after the use of the article, be reviewed and evaluated in writing by a physician who is not participating in the clinical investigation.

(c) The documentation required in paragraph (a) or (b) of this section shall be submitted to the IRB within 5 working days after the use of the test article.

2.3.1 Physical AI Adaptation of § 50.23

(d-adapted) In life-threatening situations where a Physical AI system is the only available means of delivering urgent treatment, the exception from informed consent requirements may apply to the Physical AI system component of the investigation, subject to the following additional conditions:

- (1) The Physical AI system must meet or exceed the minimum USL threshold for the procedure type (surgical: 7.0, therapeutic positioning: 6.0, diagnostic needle placement: 6.0).
- (2) Even in emergency situations, the Physical AI system must maintain its complete hash-chained audit trail. The emergency use must be recorded as an audit event with "emergency_consent_exception" classification, and the complete provenance chain from clinical data through procedure execution to outcome must be preserved.
- (3) Continuous safety monitoring must be maintained throughout the emergency procedure, including real-time force/torque limit monitoring, emergency stop availability, and qualified human oversight.
- (4) The pre-procedure safety matrix must still be completed to the extent practicable. At minimum, robot readiness verification, environmental checks, and patient identity verification must be performed. If time does not permit full safety matrix completion, the specific checks that were omitted must be documented in the audit trail.

2.3.2 Prior 21 CFR Part 50: § 50.23 Military Exception Provisions

(d) The military exception provisions of the prior 21 CFR Part 50 regulation apply in their original form to investigations involving investigational new drugs administered to members of the armed forces. When Physical AI surgical or therapeutic systems are deployed in military medical settings under these provisions, the following additional requirements apply:

- (1) The written information sheet required under 10 U.S.C. 1107(d) must include a description of any Physical AI systems that will be used in the member's treatment, including the robot type, degree of autonomy, and safety provisions.
- (2) Medical records must accurately document not only the receipt of investigational products but also all Physical AI system interactions, including robot system identifiers, software versions, procedure logs, and safety events.
- (3) DOD's recordkeeping system must be capable of tracking Physical AI system deployments and configurations, including which specific robotic platform was used for each member's procedure, with the same level of detail required for investigational drug tracking.
- (4) The IRB review required under paragraph (d)(1)(v) of the prior regulation must include evaluation of the Physical AI system's safety profile, USL score, and simulation validation evidence.

(e) The in vitro diagnostic device provisions of paragraphs (e)(1) through (e)(6) of the prior 21 CFR Part 50 regulation remain applicable in their original form. When Physical AI systems are used in conjunction with investigational in vitro diagnostic devices for identifying chemical, biological, radiological, or nuclear agents, the Physical AI system components are subject to the informed consent adaptations specified in this part.

[46 FR 8951, Jan. 27, 1981, as amended through 76 FR 36993, June 24, 2011. Physical AI adaptation added 16 March 2026.]

2.4 § 50.24 Exception from Informed Consent Requirements for Emergency Research

(a) The IRB responsible for the review, approval, and continuing review of the clinical investigation described in this section may approve that investigation without requiring that informed consent of all research subjects be obtained if the IRB (with the concurrence of a licensed physician who is a member of or consultant to the IRB and who is not otherwise participating in the clinical investigation) finds and documents each of the following:

- (1) The human subjects are in a life-threatening situation, available treatments are unproven or unsatisfactory, and the collection of valid scientific evidence, which may include evidence obtained through randomized placebo-controlled investigations, is necessary to determine the safety and effectiveness of particular interventions.
- (2) Obtaining informed consent is not feasible because: (i) the subjects will not be able to give their informed consent as a result of their medical condition; (ii) the intervention under investigation must be administered before consent from the subjects' legally authorized representatives is feasible; and (iii) there is no reasonable way to identify prospectively the individuals likely to become eligible for participation in the clinical investigation.

- (3) Participation in the research holds out the prospect of direct benefit to the subjects because: (i) subjects are facing a life-threatening situation that necessitates intervention; (ii) appropriate animal and other preclinical studies have been conducted, and the information derived from those studies and related evidence support the potential for the intervention to provide a direct benefit to the individual subjects; and (iii) risks associated with the investigation are reasonable in relation to what is known about the medical condition of the potential class of subjects, the risks and benefits of standard therapy, if any, and what is known about the risks and benefits of the proposed intervention or activity.
 - (4) The clinical investigation could not practicably be carried out without the waiver.
 - (5) The proposed investigational plan defines the length of the potential therapeutic window based on scientific evidence, and the investigator has committed to attempting to contact a legally authorized representative for each subject within that window of time and, if feasible, to asking the legally authorized representative contacted for consent within that window rather than proceeding without consent.
 - (6) The IRB has reviewed and approved informed consent procedures and an informed consent document consistent with § 50.25. These procedures and the informed consent document are to be used with subjects or their legally authorized representatives in situations where use of such procedures and documents is feasible. The IRB has reviewed and approved procedures and information to be used when providing an opportunity for a family member to object to a subject's participation in the clinical investigation consistent with paragraph (a)(7)(v) of this section.
 - (7) Additional protections of the rights and welfare of the subjects will be provided, including, at least: (i) consultation with representatives of the communities in which the clinical investigation will be conducted and from which the subjects will be drawn; (ii) public disclosure to the communities prior to initiation of the clinical investigation of plans for the investigation and its risks and expected benefits; (iii) public disclosure of sufficient information following completion of the clinical investigation to apprise the community and researchers of the study; (iv) establishment of an independent data monitoring committee to exercise oversight of the clinical investigation; and (v) if obtaining informed consent is not feasible and a legally authorized representative is not reasonably available, the investigator has committed, if feasible, to attempting to contact within the therapeutic window the subject's family member who is not a legally authorized representative, and asking whether he or she objects to the subject's participation.
- (b) The IRB is responsible for ensuring that procedures are in place to inform, at the earliest feasible opportunity, each subject, or if the subject remains incapacitated, a legally authorized representative of the subject, or if such a representative is not reasonably available, a family member, of the subject's inclusion in the clinical investigation, the details of the investigation and other information contained in the informed consent document. The IRB shall also ensure that there is a procedure to inform the subject, or if the subject remains incapacitated, a legally authorized representative, or if such a representative is not reasonably available, a family member, that he or she may discontinue the subject's participation at any time without penalty or loss of benefits to which the subject is otherwise entitled.

(c) The IRB determinations required by paragraph (a) of this section and the documentation required by paragraph (e) of this section are to be retained by the IRB for at least 3 years after completion of the clinical investigation, and the records shall be accessible for inspection and copying by FDA in accordance with § 56.115(b) of this chapter.

(d) Protocols involving an exception to the informed consent requirement under this section must be performed under a separate investigational new drug application (IND) or investigational device exemption (IDE) that clearly identifies such protocols as protocols that may include subjects who are unable to consent.

(e) If an IRB determines that it cannot approve a clinical investigation because the investigation does not meet the criteria in the exception provided under paragraph (a) of this section or because of other relevant ethical concerns, the IRB must document its findings and provide these findings promptly in writing to the clinical investigator and to the sponsor. The sponsor must promptly disclose this information to FDA and to the sponsor's clinical investigators who are participating or are asked to participate in this or a substantially equivalent investigation, and to other IRBs that have been, or are, asked to review this or a substantially equivalent investigation.

2.4.1 Physical AI Adaptation of § 50.24

(f) When emergency research under this section involves Physical AI systems, the following additional requirements apply:

- (1) Community consultation required under paragraph (a)(7)(i) must include education about Physical AI systems and their role in the emergency research protocol. Community engagement materials must explain, in accessible language, the types of robotic systems involved, the degree of autonomous operation, the safety measures in place, and the circumstances under which a Physical AI system would be used in the subject's care without prior consent.
- (2) Public disclosure required under paragraph (a)(7)(ii) must include information about the Physical AI systems to be used, including robot types, USL readiness scores, simulation validation summary, and the safety record of the platforms.
- (3) The independent data monitoring committee established under paragraph (a)(7)(iv) must include expertise in Physical AI systems or have access to such expertise through consultation. The committee's oversight responsibilities must encompass Physical AI system performance metrics, USL score trends across the investigation, safety incident patterns involving robotic systems, and the adequacy of human oversight during emergency procedures.
- (4) When family member or legally authorized representative notification occurs under paragraph (b), the notification must include information about the Physical AI system's involvement in the subject's care, including the specific robot used, the procedure performed, the degree of autonomous operation, and any Physical AI-related safety events that occurred.
- (5) The separate IND or IDE required under paragraph (d) must include Physical AI system documentation as part of the protocol, including robot capability profiles, USL assessments, simulation validation reports, and the pre-procedure safety matrix requirements applicable to emergency deployment.

[61 FR 51528, Oct. 2, 1996. Physical AI adaptation added 16 March 2026.]

2.5 § 50.25 Elements of Informed Consent

(a) *Basic elements of informed consent.* In seeking informed consent, the following information shall be provided to each subject:

- (1) A statement that the study involves research, an explanation of the purposes of the research and the expected duration of the subject's participation, a description of the procedures to be followed, and identification of any procedures which are experimental.
- (2) A description of any reasonably foreseeable risks or discomforts to the subject.
- (3) A description of any benefits to the subject or to others which may reasonably be expected from the research.
- (4) A disclosure of appropriate alternative procedures or courses of treatment, if any, that might be advantageous to the subject.
- (5) A statement describing the extent, if any, to which confidentiality of records identifying the subject will be maintained and that notes the possibility that the Food and Drug Administration may inspect the records.
- (6) For research involving more than minimal risk, an explanation as to whether any compensation and an explanation as to whether any medical treatments are available if injury occurs and, if so, what they consist of, or where further information may be obtained.
- (7) An explanation of whom to contact for answers to pertinent questions about the research and research subjects' rights, and whom to contact in the event of a research-related injury to the subject.
- (8) A statement that participation is voluntary, that refusal to participate will involve no penalty or loss of benefits to which the subject is otherwise entitled, and that the subject may discontinue participation at any time without penalty or loss of benefits to which the subject is otherwise entitled.

(b) *Additional elements of informed consent.* When appropriate, one or more of the following elements of information shall also be provided to each subject:

- (1) A statement that the particular treatment or procedure may involve risks to the subject (or to the embryo or fetus, if the subject is or may become pregnant) which are currently unforeseeable.
- (2) Anticipated circumstances under which the subject's participation may be terminated by the investigator without regard to the subject's consent.
- (3) Any additional costs to the subject that may result from participation in the research.
- (4) The consequences of a subject's decision to withdraw from the research and procedures for orderly termination of participation by the subject.
- (5) A statement that significant new findings developed during the course of the research which may relate to the subject's willingness to continue participation will be provided to the subject.
- (6) The approximate number of subjects involved in the study.

(c) When seeking informed consent for applicable clinical trials, as defined in 42 U.S.C. 282(j)(1)(A), the following statement shall be provided to each clinical trial subject in informed consent documents and processes. This will notify the clinical trial subject that clinical trial information has been or will be submitted for inclusion in the clinical trial registry databank under paragraph (j) of section 402 of the Public Health Service Act. The statement is: “A description of this clinical trial will be available on <http://www.ClinicalTrials.gov>, as required by U.S. Law. This Web site will not include information that can identify you. At most, the Web site will include a summary of the results. You can search this Web site at any time.”

(d) The informed consent requirements in these regulations are not intended to preempt any applicable Federal, State, or local laws which require additional information to be disclosed for informed consent to be legally effective.

(e) Nothing in these regulations is intended to limit the authority of a physician to provide emergency medical care to the extent the physician is permitted to do so under applicable Federal, State, or local law.

2.5.1 Physical AI Adaptation of § 50.25 Basic Elements

The eight basic elements of informed consent specified in paragraph (a) must incorporate the following Physical AI considerations when the clinical investigation involves Physical AI systems:

Element 1: Purpose and Procedures.. The statement required under paragraph (a)(1) must additionally include a clear description of the role of Physical AI systems in the investigation’s objectives. The consent document must specify which clinical procedures will involve Physical AI systems, the specific robot types by manufacturer and model designation, the degree of autonomy for each system (teleoperated, shared autonomy, or supervised autonomous), and the human oversight arrangements during each procedure. Where multiple Physical AI system types are used across different procedures within the same investigation, each system and its role must be individually described.

Element 2: Risks.. The description of risks required under paragraph (a)(2) must disclose risks specific to Physical AI system interactions, including but not limited to: mechanical risks from robotic contact (unexpected forces, unplanned instrument movements, collision events, force/torque exceedances); software risks (AI prediction errors, model confidence below clinical thresholds, software malfunction during procedures); simulation-to-reality gaps (differences between simulated robotic performance and real-world execution that may affect procedural outcomes); cybersecurity risks (potential for unauthorized access to robotic controllers, data corruption, or system compromise during clinical procedures); and emergency stop scenarios (the possibility that a procedure may be halted mid-execution due to safety limit triggers, requiring completion by conventional means).

Element 3: Benefits.. The description of benefits required under paragraph (a)(3) must describe potential advantages of Physical AI systems when supported by evidence, including but not limited to: enhanced precision and consistency of robotic procedures compared to conventional approaches; reduced human fatigue in lengthy procedures through robotic assistance; simulation-validated approaches that have been rehearsed before the actual procedure; and digital twin-guided treatment planning that uses patient-specific computational models to optimize the treatment approach.

Benefits must not be overstated, and the consent document must clearly distinguish between demonstrated benefits and theoretical advantages.

Element 4: Alternatives.. The disclosure of alternatives required under paragraph (a)(4) must explicitly include non-Physical AI alternatives for every procedure involving a Physical AI system. The subject must be informed that conventional surgical, therapeutic, or diagnostic approaches (without robotic assistance) are available as alternatives where clinically appropriate. The relative advantages and disadvantages of the Physical AI approach versus the conventional approach must be presented in a balanced manner.

Element 5: Confidentiality.. The statement on confidentiality required under paragraph (a)(5) must explain the data protections specific to Physical AI systems, including: how clinical data flows through the five MCP servers (authorization, FHIR clinical data, DICOM imaging, audit ledger, provenance tracking); that all data provided to Physical AI systems undergoes HIPAA Safe Harbor de-identification with removal or generalization of all 18 PHI identifiers; that HMAC-SHA256 pseudonymization with site-specific salts is used when re-linkable data is necessary; that all Physical AI system data access operates under deny-by-default authorization where access is denied unless explicitly permitted; that every data access event by Physical AI systems is recorded in a hash-chained immutable audit trail satisfying 21 CFR Part 11; and that in multi-site investigations using federated learning, differential privacy mechanisms (epsilon and delta parameters) and secure aggregation protocols protect individual patient data from exposure across sites.

Element 6: Compensation and Treatment for Injury.. The explanation required under paragraph (a)(6) must address liability and compensation in cases where Physical AI system malfunction causes injury to the subject. The consent document must specify: whether the sponsor, the robotic system manufacturer, or the AI model developer bears responsibility for injuries arising from Physical AI system interactions; what medical treatments are available if a Physical AI system-related injury occurs, including injuries from mechanical contact, AI-driven treatment modifications, or device malfunction; and how to report suspected Physical AI system-related injuries.

Element 7: Contact Information.. The contact information required under paragraph (a)(7) must include, in addition to standard investigator contacts: a designated contact for Physical AI system safety concerns who has expertise in the robotic systems, AI models, or digital twin technologies used in the investigation; and a contact for cybersecurity concerns related to Physical AI systems processing the subject's data.

Element 8: Voluntary Participation.. The statement on voluntary participation required under paragraph (a)(8) must explicitly state that the subject may decline Physical AI system interactions for any specific procedure and request a non-Physical AI alternative at any time during the investigation, without penalty or loss of benefits, where such alternatives exist and are clinically appropriate.

2.5.2 Physical AI Additional Elements of Informed Consent

In addition to the basic and additional elements specified in paragraphs (a) and (b), the following elements must be provided to each subject in clinical investigations involving Physical AI systems:

- (1) *USL Readiness Score.* The USL readiness score of the robotic system assigned to the subject's procedure, expressed as a numerical score (1.0 to 10.0) with plain-language explanation of the corresponding level (Minimal through Reference) and readiness band (Foundational, Intermediate, or Advanced). The minimum USL threshold required for the procedure type must be stated alongside the assigned platform's actual score, presented in a format accessible to a lay audience.
- (2) *Simulation Rehearsal.* Whether the specific procedure planned for the subject was rehearsed in simulation prior to the clinical procedure, which simulation framework(s) were used (NVIDIA Isaac Lab v2.3.1, MuJoCo v3.4.0, Gazebo Sim v10.0.0, or PyBullet v3.2.5), and the cross-framework validation results demonstrating behavioral consistency. If simulation rehearsal was not performed, the reason must be stated.
- (3) *Digital Twin.* Whether a digital twin computational model was or will be created from the subject's clinical data, what patient data was used to construct the model (imaging, biomarkers, clinical measurements), how the digital twin will be used (treatment planning, procedure guidance, response prediction), whether the digital twin model will be retained after the investigation concludes, and the subject's right to request deletion of their digital twin model.
- (4) *Robot Agent Workflow.* A plain-language description of the six-step robot agent workflow that will govern the procedure, as defined in the national MCP-PAI standard: (Step 1) the robot authenticates with the authorization server and receives a scoped access token; (Step 2) the robot retrieves the subject's de-identified clinical data through the FHIR server; (Step 3) the robot accesses the subject's de-identified imaging data through the DICOM server; (Step 4) the robot executes the clinical procedure under human oversight; (Step 5) the robot records a complete audit trail of all actions taken; and (Step 6) the robot records the data provenance chain documenting how the subject's data was used throughout the procedure.
- (5) *Pre-Procedure Safety Matrix.* A description of the safety verification checks that must pass before the Physical AI system may begin the subject's procedure: valid authorization, patient identity verification, clinical data availability, imaging data access confirmation, robot readiness and USL threshold verification, environmental safety checks, simulation validation confirmation, and digital twin synchronization (where applicable). The subject must be informed that the procedure will not proceed if any required safety check fails.
- (6) *Right to Non-Physical AI Alternative.* A clear statement of the subject's right to request a non-Physical AI alternative for any procedure at any time during the investigation, including after initial consent has been given. The statement must specify what conventional alternatives are available and confirm that choosing a conventional alternative will not affect the subject's eligibility for other aspects of the investigation or their access to standard-of-care treatment.
- (7) *Real-Time Safety Monitoring.* A description of how real-time safety monitoring will be conducted during the procedure, including: the identity and qualifications of the clinical personnel maintaining supervisory authority; the continuous authorization verification performed by the MCP authorization server; the force/torque limit monitoring active throughout the procedure; the emergency stop capability and how it would be activated; and the immediate recording of all safety events in the hash-chained audit trail.

- (8) *Post-Procedure Provenance.* A description of the post-procedure documentation that will be generated, including: the complete audit trail of all Physical AI system actions during the procedure; the data provenance chain documenting the lineage from the subject's clinical data through procedure execution to outcomes recording; the safety incident report (if any safety events occurred); and the USL score update reflecting the procedure outcome.

[46 FR 8951, Jan. 27, 1981, as amended at 76 FR 270, Jan. 4, 2011. Physical AI adaptation added 16 March 2026.]

2.6 § 50.27 Documentation of Informed Consent

(a) Except as provided in § 56.109(c), informed consent shall be documented by the use of a written consent form approved by the IRB and signed and dated by the subject or the subject's legally authorized representative at the time of consent. A copy shall be given to the person signing the form.

(b) Except as provided in § 56.109(c), the consent form may be either of the following:

- (1) A written consent document that embodies the elements of informed consent required by § 50.25. This form may be read to the subject or the subject's legally authorized representative, but, in any event, the investigator shall give either the subject or the representative adequate opportunity to read it before it is signed.
- (2) A *short form* written consent document stating that the elements of informed consent required by § 50.25 have been presented orally to the subject or the subject's legally authorized representative. When this method is used, there shall be a witness to the oral presentation. Also, the IRB shall approve a written summary of what is to be said to the subject or the representative. Only the short form itself is to be signed by the subject or the representative. However, the witness shall sign both the short form and a copy of the summary, and the person actually obtaining the consent shall sign a copy of the summary. A copy of the summary shall be given to the subject or the representative in addition to a copy of the short form.

2.6.1 Physical AI Adaptation of § 50.27

(c) When the written consent document or short form is used for clinical investigations involving Physical AI systems, the following additional documentation requirements apply:

- (1) The written consent document must include a Physical AI System Summary section that consolidates the Physical AI-specific elements required under § 50.25 into a clearly identified portion of the consent form. This section must be written in plain language accessible to the subject population and must include visual aids, photographs, or diagrams of the robotic systems where practical.
- (2) When the short form method is used, the oral presentation must include a demonstration or visual explanation of the Physical AI systems the subject may encounter. The written summary approved by the IRB must include all Physical AI-specific elements from § 50.25.
- (3) The consent process must be recorded as an audit event in the MCP ledger server, documenting the date and time of consent, the identity of the person obtaining consent, the version of the consent document used, and the specific Physical AI system configurations disclosed. The consent status must be tracked using the consent-status schema defined in the national MCP-PAI standard, which supports six granular consent categories: general trial participation,

Physical AI procedure consent, data processing consent, digital twin consent, federated learning consent, and biospecimen consent.

- (4) Any subsequent modifications to the Physical AI systems disclosed in the original consent document (including software updates, AI model version changes, or substitution of a different robotic platform) must trigger a re-consent evaluation by the IRB as specified in § 50.32 of this part.

[46 FR 8951, Jan. 27, 1981, as amended at 61 FR 57280, Nov. 5, 1996. Physical AI adaptation added 16 March 2026.]

3 SUBPART C – ADDITIONAL PROTECTIONS FOR SUBJECTS IN PHYSICAL AI CLINICAL INVESTIGATIONS

This Subpart C establishes additional protections for human subjects in clinical investigations that involve Physical AI systems. The prior 21 CFR Part 50 reserved Subpart C. This adaptation utilizes Subpart C to address protections specific to Physical AI systems that do not fit naturally into the existing Subpart A (General Provisions) or Subpart B (Informed Consent) sections. These protections are derived from the physical-ai-oncology-trials repository (v2.3.0, DOI: 10.5281/zenodo.18445179), the national MCP-PAI oncology trials standard (v1.2.0, DOI: 10.5281/zenodo.18869776), the ICH E6(R3) Physical AI adaptation (DOI: 10.5281/zenodo.18973368), and the federated learning framework (DOI: 10.5281/zenodo.18840880).

3.1 § 50.30 Physical AI System Safety Requirements

3.1.1 Pre-Procedure Safety Matrix

(a) Before any Physical AI system may begin a clinical procedure on a human subject, a pre-procedure safety matrix must be completed and documented. The safety matrix consists of the following verification checks, all of which must pass before the procedure may proceed:

- (1) *Authorization.* A valid authorization token must be issued by the `trialmcp-authz` server, with scope appropriate to the planned procedure type, the assigned robot agent, and the specific subject. The token must not be expired and must not have been revoked.
- (2) *Patient Identity Verification.* The subject's identity must be verified against the clinical record accessed through the `trialmcp-fhir` server. The verification must confirm that the subject enrolled in the investigation is the same individual presenting for the procedure.
- (3) *Clinical Data Availability.* The subject's relevant clinical data must be accessible through the `trialmcp-fhir` server, including current diagnosis, treatment history, allergies, relevant laboratory results, and any contraindications for robotic procedures. Data must be de-identified per HIPAA Safe Harbor before delivery to the Physical AI system.
- (4) *Imaging Data Access.* Where the procedure requires imaging guidance, the subject's relevant imaging data must be accessible through the `trialmcp-dicom` server with appropriate role-based modality restrictions (CT, MR, PT as applicable). Imaging metadata must be normalized and patient identifiers hashed per the DICOM integration specifications.
- (5) *Robot Readiness and USL Threshold.* The Physical AI system must be verified as operationally ready, with current calibration, software version matching the protocol specification, and

all safety systems functional. The platform's current USL score must meet or exceed the minimum threshold for the procedure type: surgical robots require USL 7.0 or above, therapeutic positioning and diagnostic needle placement require USL 6.0 or above, rehabilitative exoskeletons require USL 4.0 or above, and companion monitoring systems require USL 3.0 or above.

- (6) *Environmental Checks.* The clinical environment must meet the Physical AI system's operational requirements, including adequate space, appropriate lighting, electromagnetic compatibility, network connectivity for MCP server communication, and absence of interference sources that could affect robotic control or sensor performance.
- (7) *Simulation Validation.* The Physical AI system must have undergone simulation validation for the planned procedure type, with documented cross-framework behavioral consistency demonstrating trajectory deviations below 2mm and force discrepancies below 0.5N across at least two recognized simulation frameworks.
- (8) *Digital Twin Synchronization.* Where a digital twin model is used for procedure planning or guidance, the twin must be synchronized with the subject's most recent clinical data, calibrated to acceptable accuracy thresholds, and verified as computationally current before the procedure begins.

3.1.2 Runtime Safety Monitoring

(b) During the execution of any clinical procedure involving a Physical AI system, the following runtime safety monitoring requirements must be maintained continuously:

- (1) *Continuous Authorization.* The Physical AI system's authorization token must remain valid throughout the procedure. The `trialmcp-authz` server must be queried at regular intervals (not exceeding 60 seconds) to verify that the token has not been revoked and that no policy changes have occurred that would affect the system's permissions.
- (2) *Real-Time Audit Recording.* Every significant action taken by the Physical AI system during the procedure must be recorded in the hash-chained audit trail via the `trialmcp-ledger` server. Audit records must include the action type, timestamp, actor identity, resource affected, outcome, and a SHA-256 hash linking the record to the previous entry in the chain.
- (3) *Emergency Stop Availability.* An emergency stop capability must be immediately accessible to the supervising clinical personnel at all times during the procedure. Activation of the emergency stop must immediately halt all Physical AI system motion, preserve the current state for post-incident analysis, record an emergency stop audit event, and transition the task order to a failed state.
- (4) *Force and Torque Limit Monitoring.* For Physical AI systems that make physical contact with the subject, continuous monitoring of applied forces and torques must be maintained. The monitoring system must detect any exceedance of the pre-specified safety limits and must trigger an immediate protective response (speed reduction, controlled stop, or emergency stop) when limits are approached or exceeded.
- (5) *Human Oversight.* A qualified clinical professional with authority to intervene, override, or terminate the Physical AI system's operations must be physically present and attentive

throughout the procedure. The supervising professional must have received documented training on the specific Physical AI system and must have access to override controls.

3.1.3 Post-Procedure Requirements

(c) Following the completion (or termination) of any clinical procedure involving a Physical AI system, the following post-procedure requirements must be fulfilled:

- (1) *Complete Audit Trail.* The complete hash-chained audit trail for the procedure must be finalized and verified for chain integrity via the trialmcp-ledger server. The final audit record must document the procedure outcome (completed, cancelled, or failed), the total procedure duration, and any safety events that occurred.
- (2) *Provenance Chain.* A complete data provenance record must be created via the trialmcp-provenance server, documenting the directed acyclic graph (DAG) of data lineage from the subject's clinical data through all processing steps, procedure execution, and outcome recording. Each node in the provenance DAG must include a SHA-256 fingerprint for integrity verification.
- (3) *Outcome Recording.* The clinical outcome of the procedure must be recorded in a structured format, linked to the audit trail and provenance chain, and made available for monitoring and safety review.
- (4) *Safety Incident Reporting.* Any safety events that occurred during the procedure (emergency stop activations, force limit exceedances, software anomalies, unexpected AI behaviors, equipment malfunctions) must be documented in a structured safety incident report, linked to the relevant audit records, and reported to the sponsor and IRB in accordance with the investigation's safety reporting procedures.
- (5) *USL Score Update.* The procedure outcome must be incorporated into the ongoing USL assessment for the Physical AI platform, contributing to the clinical trial collaboration dimension (Dimension 4) of the USL score. Aggregate procedure outcomes across the investigation inform the platform's evolving readiness profile.

3.1.4 Task-Order Lifecycle

(d) Every clinical procedure involving a Physical AI system must be governed by a task order that progresses through the following defined lifecycle states:

- (1) *Scheduled.* The procedure has been scheduled and a task order created, specifying the procedure type, assigned robot platform, subject identifier (pseudonymized), scheduled time, and required safety checks.
- (2) *Safety Check.* The pre-procedure safety matrix specified in paragraph (a) is being executed. The task order remains in this state until all required safety checks pass or until a check failure terminates the sequence.
- (3) *In Progress.* All safety checks have passed and the Physical AI system is actively executing the clinical procedure. Runtime safety monitoring specified in paragraph (b) is active throughout this state.
- (4) *Completed.* The procedure has been successfully completed and all post-procedure requirements specified in paragraph (c) have been fulfilled.

- (5) *Cancelled.* The procedure was cancelled before execution began (e.g., due to safety check failure, subject withdrawal, or scheduling change).
- (6) *Failed.* The procedure was terminated during execution due to a safety event, emergency stop activation, equipment malfunction, or clinical decision to abort. All post-procedure requirements must still be fulfilled for failed procedures.

3.1.5 Forbidden Operations

(e) The following operations are strictly prohibited for Physical AI systems operating in clinical investigations:

- (1) No Physical AI system may begin a clinical procedure without completing the pre-procedure safety matrix specified in paragraph (a).
- (2) No Physical AI system may operate on a subject without a valid, non-expired, non-revoked authorization token from the trialmcp-authz server.
- (3) No Physical AI system may skip or disable audit recording during any clinical procedure. Every procedure must produce a complete, hash-chained audit trail.
- (4) No Physical AI system may operate in a clinical investigation if its current USL score falls below the minimum threshold specified for the procedure type.
- (5) No Physical AI system may directly modify clinical systems (EHR, PACS, laboratory information systems) without explicit authorization mediated through the appropriate MCP server. All clinical system interactions must be audited and traceable.
- (6) No Physical AI system may continue a clinical procedure after an emergency stop has been activated without completing a new full safety check cycle as specified in paragraph (a). The original task order must be marked as failed, and a new task order must be created for any resumption of the procedure.

3.2 § 50.31 IRB Review of Physical AI Investigations

(a) In addition to the IRB review requirements specified elsewhere in this chapter, IRBs reviewing clinical investigations that involve Physical AI systems must evaluate the following:

- (1) *Physical AI System Safety Profiles.* The IRB must review the safety profile of each Physical AI system specified in the protocol, including mechanical safety data, software reliability evidence, AI model validation results, simulation testing outcomes, and any prior clinical experience with the system.
- (2) *USL Scores.* The IRB must review the USL readiness score for each robotic platform and verify that the score meets or exceeds the minimum threshold for the proposed procedure type. The IRB must assess whether the four-dimension USL evaluation (simulation switching, AI integration, cross-robot sharing, clinical trial collaboration) adequately characterizes the platform's readiness for the planned clinical use.
- (3) *Simulation Validation Reports.* The IRB must review simulation validation evidence demonstrating that the Physical AI system has been tested in at least two recognized simulation frameworks with documented cross-framework behavioral consistency within acceptable tolerances (trajectory deviation below 2mm, force discrepancy below 0.5N).

- (4) *Human Oversight Plans.* The IRB must evaluate the adequacy of the human oversight plan for the investigation, including the qualifications of supervising personnel, the availability and accessibility of override controls, the emergency stop provisions, and the procedures for managing Physical AI system failures during clinical procedures.
 - (5) *Informed Consent Adequacy.* The IRB must assess whether the informed consent document adequately addresses all Physical AI-specific elements required under §§ 50.20 and 50.25 of this part, including the disclosure of robot types, USL scores, simulation validation, digital twin usage, MCP data protections, and the subject's right to non-Physical AI alternatives.
 - (6) *Robot Capability Profiles.* The IRB must review the machine-readable robot capability profile for each Physical AI platform, as defined in the national MCP-PAI standard, and verify that the platform's documented capabilities, safety prerequisites, and required MCP tools are appropriate for the proposed clinical procedures.
- (b) IRBs must have access to expertise in Physical AI systems sufficient to evaluate protocols involving robotic systems, AI/ML models, digital twins, and simulation technologies. This expertise may be provided through IRB membership, consultation with qualified experts, or collaboration with institutions that have established Physical AI review capabilities.
- (c) The IRB site policy template from the national MCP-PAI standard provides a reference framework for developing institution-specific IRB review procedures for Physical AI investigations. Institutions should adapt this template to their local governance requirements.

3.3 § 50.32 Ongoing Consent and Subject Notification

- (a) Subjects enrolled in clinical investigations involving Physical AI systems must be notified of significant changes to the Physical AI systems used in their care. Significant changes include, but are not limited to:
- (1) Software updates to robotic control systems or AI models that alter the system's behavior, capabilities, or safety characteristics.
 - (2) AI model version changes where the updated model was trained on different data, uses a modified architecture, or produces materially different outputs than the version disclosed in the original consent.
 - (3) Changes to the platform's USL score that move the score below the minimum threshold for the procedure type or that change the platform's readiness band classification.
 - (4) Substitution of a different robotic platform than the one originally disclosed in the informed consent document, even if the substitute platform is of the same type and meets the USL threshold.
 - (5) Changes to the MCP server configuration, security policies, or data protection mechanisms that affect how the subject's data is processed, stored, or shared.
- (b) When significant changes are identified, the investigator must notify the IRB and determine whether re-consent is required. Changes governed by a Predetermined Change Control Plan (PCCP) that have been pre-approved by the IRB may be implemented with notification rather than re-consent, provided the PCCP was disclosed in the original consent document and the subject consented to the PCCP-governed update process.

(c) Subjects must be informed of their right to withdraw consent for Physical AI system interactions at any time, including after learning of a significant change. Withdrawal of consent for Physical AI interactions does not require withdrawal from the overall investigation if non-Physical AI alternatives are available.

3.4 § 50.33 Data Protection for Physical AI Investigations

(a) *De-Identification.* All clinical data accessed by Physical AI systems must be de-identified per HIPAA Safe Harbor requirements before delivery to the Physical AI system. De-identification must remove or generalize all 18 categories of Protected Health Information as defined in 45 CFR 164.514(b)(2): names; geographic subdivisions smaller than a state; dates (except year) related to the individual; telephone numbers; fax numbers; email addresses; Social Security numbers; medical record numbers; health plan beneficiary numbers; account numbers; certificate/license numbers; vehicle identifiers and serial numbers; device identifiers and serial numbers; web universal resource locators; internet protocol addresses; biometric identifiers; full-face photographs and comparable images; and any other unique identifying number, characteristic, or code.

(b) *Pseudonymization.* When re-linkable data is required for longitudinal tracking within the investigation, pseudonymization must use HMAC-SHA256 with site-specific salts. The pseudonymization key material must be stored separately from the pseudonymized data, with access restricted to authorized personnel at the clinical site. Physical AI systems must not have access to the pseudonymization key material.

(c) *Deny-by-Default Authorization.* Physical AI systems must operate under deny-by-default role-based access control (RBAC) as enforced by the trialmcp-authz server. All data access requests must be explicitly authorized by policy before being granted. The six-actor permission matrix defined in the national MCP-PAI standard governs access: Robot Agent, Trial Coordinator, Data Monitor, Auditor, Sponsor, and CRO each have explicitly defined permissions. Explicit DENY rules take precedence over ALLOW rules in all cases.

(d) *Hash-Chained Audit Trails.* All Physical AI system actions must be recorded in hash-chained immutable audit trails via the trialmcp-ledger server. Each audit record must contain the action performed, actor identity, timestamp (UTC), resource affected, outcome, and a SHA-256 hash computed over the canonical JSON representation of the record concatenated with the hash of the previous record. The genesis record uses a defined genesis hash. These audit trails satisfy the electronic records requirements of 21 CFR Part 11 for attributability, legibility, contemporaneousness, originality, and accuracy.

(e) *Provenance Tracking.* The complete data lineage for every Physical AI procedure must be documented through DAG-based provenance tracking via the trialmcp-provenance server. Provenance records must trace the path from source clinical data (FHIR resources, DICOM studies) through all transformations, processing steps, and analytical operations to final clinical outcomes. Each provenance node must include a SHA-256 fingerprint of the data at that point in the lineage, enabling verification of data integrity throughout the processing chain.

(f) *Federated Learning.* In multi-site investigations that employ federated learning for AI model training, the following data protections must be implemented: differential privacy mechanisms must be applied to model updates shared across sites, with documented epsilon and delta parameters; secure aggregation protocols (additive secret sharing, pairwise masking) must prevent any single

site or aggregator from accessing another site's raw model updates; data harmonization across sites must be documented, including mappings between DICOM metadata formats, FHIR resource representations, ICD-10 to SNOMED CT coding, and LOINC tumor marker standardization; and privacy budget tracking must ensure that cumulative privacy expenditure across all federated learning rounds does not exceed the pre-specified privacy budget for the investigation.

3.5 § 50.34 Physical AI System Classification and Regulatory Pathways

(a) *FDA AI/ML SaMD Classification*. Physical AI systems that incorporate AI/ML components functioning as Software as a Medical Device (SaMD) are subject to FDA regulatory classification and oversight. The classification of the AI/ML component depends on the significance of the information provided by the software to the healthcare decision and the state of the healthcare situation or condition, consistent with the International Medical Device Regulators Forum (IMDRF) risk categorization framework and FDA guidance on AI/ML-enabled medical devices (January 2025).

(b) *Regulatory Submission Pathways*. Depending on the risk classification of the Physical AI system, the following regulatory submission pathways may apply:

- (1) *510(k) Premarket Notification*. For Physical AI systems that are substantially equivalent to a legally marketed predicate device. This pathway may be appropriate for incremental AI/ML enhancements to existing robotic surgical platforms with established predicates.
- (2) *De Novo Classification*. For novel Physical AI systems of low to moderate risk that lack a suitable predicate. This pathway may be appropriate for novel AI-guided diagnostic or therapeutic robotic systems that represent new device types.
- (3) *Premarket Approval (PMA)*. For high-risk Physical AI systems, including autonomous surgical robots and AI systems that independently make critical clinical decisions. PMA requires demonstration of safety and effectiveness through valid scientific evidence.
- (4) *Breakthrough Device Designation*. Physical AI systems that provide more effective treatment or diagnosis of life-threatening or irreversibly debilitating diseases may qualify for Breakthrough Device Designation, which provides intensive FDA interaction and priority review.

(c) *Predetermined Change Control Plan (PCCP)*. Adaptive Physical AI systems that are designed to learn and evolve based on clinical experience must maintain a documented PCCP that specifies: the types of modifications anticipated (e.g., AI model retraining, parameter updates, algorithm refinements); the methodology for implementing and validating each type of change; the performance metrics and acceptance criteria that modifications must satisfy before deployment; and the process for assessing the impact of modifications on safety and effectiveness.

(d) *Applicable Safety Standards*. Physical AI systems used in clinical investigations must comply with applicable safety standards, including but not limited to:

- (1) IEC 80601-2-77 for robotically assisted surgical equipment, which addresses basic safety and essential performance requirements for surgical robotic systems.
- (2) ISO 14971 for application of risk management to medical devices, which provides the framework for identifying, evaluating, controlling, and monitoring risks associated with Physical AI systems throughout their lifecycle.

- (3) ISO 13482 for safety requirements for personal care robots, which is applicable to rehabilitative exoskeletons, companion robots, and other Physical AI systems that interact directly with patients in non-surgical capacities.
- (4) IEC 62304 for medical device software lifecycle processes, which governs the development, maintenance, and risk management of software components within Physical AI systems.
- (5) ISO/TS 15066 for collaborative robot safety, which specifies safety requirements for cobots operating alongside clinical staff in shared workspaces.

4 SUBPART D – ADDITIONAL SAFEGUARDS FOR CHILDREN IN CLINICAL INVESTIGATIONS

Source: 66 FR 20598, Apr. 24, 2001, unless otherwise noted.

4.1 § 50.50 IRB Duties

In addition to other responsibilities assigned to IRBs under this subpart, each IRB must review proposed clinical investigations involving children as subjects covered by this subpart and approve only those clinical investigations that satisfy the criteria described in § 50.51, § 50.52, or § 50.53, or meet the conditions described in § 50.54.

4.1.1 Physical AI Adaptation of § 50.50

When the clinical investigation involves Physical AI systems and children as subjects, the IRB must additionally evaluate whether the Physical AI system's safety profile, human oversight plan, and informed consent materials are appropriate for the pediatric population. The IRB must consider the child's age, developmental stage, and ability to understand and interact with robotic systems when evaluating the adequacy of assent procedures and Physical AI system configurations. For assistive robots used in pediatric oncology settings (companion robots, humanoid assistants), the IRB must evaluate the age-appropriateness of the robotic interaction protocols.

4.2 § 50.51 Clinical Investigations Not Involving Greater Than Minimal Risk

An IRB may find that a clinical investigation involving children as subjects involves no greater than minimal risk, provided that adequate provisions are made for soliciting the assent of the children and the permission of their parents or guardians as set forth in § 50.55.

4.2.1 Physical AI Adaptation of § 50.51

When Physical AI systems are involved in clinical investigations with children that are proposed as involving no greater than minimal risk, the IRB must apply the Physical AI minimal risk guidelines specified in § 50.22(f) of this part. Companion monitoring systems performing only non-contact observation and verbal interaction may qualify for minimal risk classification in pediatric settings if the system's interaction protocols have been specifically validated for the target age range. Robotic systems involving physical contact with children generally do not qualify for minimal risk classification.

4.3 § 50.52 Clinical Investigations Involving Greater Than Minimal Risk but Presenting the Prospect of Direct Benefit

An IRB may find that a clinical investigation involving children as subjects involves greater than minimal risk but presents the prospect of direct benefit to the individual subjects, provided that:

- (a) The risk is justified by the anticipated benefit to the subjects;
- (b) The relation of the anticipated benefit to the risk is at least as favorable to the subjects as that presented by available alternative approaches; and
- (c) Adequate provisions are made for soliciting the assent of the children and permission of their parents or guardians as set forth in § 50.55.

4.3.1 Physical AI Adaptation of § 50.52

When Physical AI systems are used in investigations involving children with the prospect of direct benefit, the risk-benefit assessment required under paragraph (a) must specifically account for Physical AI-specific risks (mechanical injury, AI prediction error, software malfunction) and Physical AI-specific benefits (enhanced surgical precision, reduced procedure duration, simulation-validated approaches). The comparison with alternative approaches required under paragraph (b) must include comparison with the same procedure performed without Physical AI assistance.

4.4 § 50.53 Clinical Investigations Involving Greater Than Minimal Risk and No Prospect of Direct Benefit

An IRB may find that a clinical investigation involving children as subjects involves greater than minimal risk and no prospect of direct benefit to the individual subjects, provided that:

- (a) The risk represents a minor increase over minimal risk;
- (b) The intervention or procedure presents experiences to subjects that are reasonably commensurate with those inherent in their actual or expected medical, dental, psychological, social, or educational situations;
- (c) The intervention or procedure is likely to yield generalizable knowledge about the subjects' disorder or condition that is of vital importance for the understanding or amelioration of the subjects' disorder or condition; and
- (d) Adequate provisions are made for soliciting the assent of the children and permission of their parents or guardians as set forth in § 50.55.

4.4.1 Physical AI Adaptation of § 50.53

Physical AI systems involving physical contact with children should generally not be approved under this section unless the robotic interaction is limited to monitoring or rehabilitative functions that represent only a minor increase over minimal risk. The use of surgical robots, therapeutic positioning systems, or diagnostic needle-placement platforms on children without prospect of direct benefit to the individual child is not appropriate under this section.

4.5 § 50.54 Clinical Investigations Not Otherwise Approvable

- (a) The IRB finds that the clinical investigation presents a reasonable opportunity to further the understanding, prevention, or alleviation of a serious problem affecting the health or welfare of children; and
- (b) The Commissioner of Food and Drugs, after consultation with a panel of experts in pertinent disciplines and following opportunity for public review and comment, determines either that the investigation satisfies the conditions of § 50.51, § 50.52, or § 50.53, or that the investigation presents a reasonable opportunity to further the understanding, prevention, or alleviation of a serious problem

affecting the health or welfare of children, the investigation will be conducted in accordance with sound ethical principles, and adequate provisions are made for soliciting the assent of children and the permission of their parents or guardians as set forth in § 50.55.

4.5.1 Physical AI Adaptation of § 50.54

When Physical AI systems are involved in investigations submitted to the Commissioner under this section, the panel of experts consulted must include at least one member with expertise in Physical AI systems, pediatric robotic surgery, or pediatric medical device safety. The public review and comment period must include materials explaining the Physical AI systems proposed for use with children in accessible, non-technical language.

4.6 § 50.55 Requirements for Permission by Parents or Guardians and for Assent by Children

- (a) In addition to the determinations required under other applicable sections of this subpart D, the IRB must determine that adequate provisions are made for soliciting the assent of the children when in the judgment of the IRB the children are capable of providing assent.
- (b) In determining whether children are capable of providing assent, the IRB must take into account the ages, maturity, and psychological state of the children involved. This judgment may be made for all children to be involved in clinical investigations under a particular protocol, or for each child, as the IRB deems appropriate.
- (c) The assent of the children is not a necessary condition for proceeding with the clinical investigation if the IRB determines that the capability of some or all of the children is so limited that they cannot reasonably be consulted, or that the intervention or procedure involved in the clinical investigation holds out a prospect of direct benefit that is important to the health or well-being of the children and is available only in the context of the clinical investigation.
- (d) Even where the IRB determines that the subjects are capable of assenting, the IRB may still waive the assent requirement if it finds and documents that the clinical investigation involves no more than minimal risk, the waiver will not adversely affect the rights and welfare of the subjects, the investigation could not practicably be carried out without the waiver, and whenever appropriate, the subjects will be provided with additional pertinent information after participation.
- (e) In addition to the determinations required under other applicable sections of this subpart D, the IRB must determine, in accordance with and to the extent that consent is required under part 50, that the permission of each child's parents or guardian is granted. Where parental permission is to be obtained, the IRB may find that the permission of one parent is sufficient for investigations conducted under § 50.51 or § 50.52. Where investigations are covered by § 50.53 or § 50.54 and permission is to be obtained from parents, both parents must give their permission unless one parent is deceased, unknown, incompetent, or not reasonably available, or when only one parent has legal responsibility for the care and custody of the child.
- (f) Permission by parents or guardians must be documented in accordance with and to the extent required by § 50.27.
- (g) When the IRB determines that assent is required, it must also determine whether and how assent must be documented.

4.6.1 Physical AI Adaptation of § 50.55

(h) When clinical investigations involve Physical AI systems and children as subjects, the assent process must include age-appropriate explanations of the robotic systems the child may encounter. For younger children who may not understand technical descriptions, visual aids including photographs, videos, or supervised demonstrations of the robotic system in a non-clinical setting should be provided. For older children and adolescents capable of understanding more detailed information, the assent materials should include a simplified version of the Physical AI-specific consent elements required under § 50.25 of this part.

(i) Parental permission for Physical AI investigations must include all Physical AI-specific elements required in § 50.25, including disclosure of the specific robot type, USL score, safety provisions, and the parent's right to request non-Physical AI alternatives for their child.

4.7 § 50.56 Wards

(a) Children who are wards of the State or any other agency, institution, or entity can be included in clinical investigations approved under § 50.53 or § 50.54 only if such clinical investigations are related to their status as wards, or conducted in schools, camps, hospitals, institutions, or similar settings in which the majority of children involved as subjects are not wards.

(b) If the clinical investigation is approved under paragraph (a), the IRB must require appointment of an advocate for each child who is a ward. The advocate will serve in addition to any other individual acting on behalf of the child as guardian or in loco parentis. One individual may serve as advocate for more than one child. The advocate must be an individual who has the background and experience to act in, and agrees to act in, the best interest of the child for the duration of the child's participation in the clinical investigation. The advocate must not be associated in any way (except in the role as advocate or member of the IRB) with the clinical investigation, the investigator(s), or the guardian organization.

4.7.1 Physical AI Adaptation of § 50.56

(c) When clinical investigations under this section involve Physical AI systems, the advocate appointed for each ward must be informed about the Physical AI systems used in the investigation, including the types of robotic systems, the degree of autonomous operation, the safety provisions, and the availability of non-Physical AI alternatives. The advocate must have access to the Physical AI-specific consent elements described in § 50.25 and must be given adequate time and opportunity to understand the Physical AI aspects of the investigation before the ward's participation begins.

5 GLOSSARY

The following definitions supplement the standard regulatory definitions in § 50.3 with terms specific to Physical AI oncology clinical investigations. Standard 21 CFR Part 50 definitions (Act, Application for research or marketing permit, Clinical investigation, Investigator, Sponsor, Human subject, Institution, IRB, Test article, Minimal risk, Legally authorized representative, Family member, Assent, Children, Parent, Ward, Permission, Guardian) retain their definitions from § 50.3 and are not repeated here.

Agentic AI: Artificial intelligence systems capable of autonomous reasoning, planning, and action execution with minimal human prompting. In oncology clinical trials, agentic AI includes LLM-based surgical assistants, multi-agent coordination systems (CrewAI v1.6.1, LangGraph v1.1.0), and autonomous simulation orchestrators operating under human oversight.

AI Model Drift: The degradation of AI model performance over time due to changes in the input data distribution, population characteristics, or clinical context. Detection and management of drift is a critical ongoing monitoring activity in Physical AI investigations.

Calibration (Robotic): The process of measuring and adjusting a robotic system’s sensors, actuators, and coordinate systems to ensure accurate physical operation. Calibration verification should be performed before each clinical use session as part of the pre-procedure safety matrix.

Calibration (AI Model): The degree to which an AI model’s predicted probabilities match the observed frequencies of outcomes. Well-calibrated models produce predictions that are neither systematically overconfident nor underconfident.

Cobot (Collaborative Robot): A robotic system designed to operate alongside human workers in a shared workspace, equipped with force limiting, speed reduction, and collision detection. Evaluated cobots: Franka Emika Panda (USL 7.4), Kinova Gen3 (USL 5.7), UFACTORY xArm 7 (USL 3.4).

Conformance Level: One of five cumulative levels defined in the national MCP-PAI standard: Core (Level 1), Clinical Read (Level 2), Imaging (Level 3), Federated Site (Level 4), and Robot Procedure (Level 5). Each level builds upon the requirements of the previous level.

Cross-Framework Validation: Verification that robotic behaviors trained in one simulation framework produce equivalent results in a different framework. The unification bridge (v1.0.0) enables validation across NVIDIA Isaac Lab, MuJoCo, Gazebo, and PyBullet.

Cybersecurity Incident: An event that compromises the confidentiality, integrity, or availability of a Physical AI system or its data, including unauthorized access, parameter tampering, data corruption, and network-based attacks.

Differential Privacy: A mathematical framework for quantifying and limiting privacy loss when publishing statistical analyses. In federated oncology trials, epsilon and delta parameters govern the noise added to gradient updates and aggregate statistics.

Digital Twin: A patient-specific computational model continuously updated with real-world data to provide predictions, simulations, and decision support. The digital twins framework (v1.0.0 through v1.3.0) supports tumor modeling, treatment simulation, and clinical integration via FHIR/DICOM.

Diffusion Policy: A generative AI technique using denoising diffusion probabilistic models to generate robotic trajectories for complex manipulation tasks.

Domain Randomization: A simulation training technique that varies environmental parameters during training to produce robotic policies that transfer more reliably from simulation to real-world deployment.

Federated Learning: Multi-site AI model training without centralizing raw data, using FedAvg, FedProx, or SCAFFOLD algorithms with differential privacy and secure aggregation (DOI: 10.5281/zenodo.18840880).

Force Limiting: A safety mechanism restricting the maximum force a robot can exert on a person or object, critical for clinical robotic procedures and validated throughout the investigation.

Hash-Chained Audit Trail: An immutable, tamper-evident record using SHA-256 hash chains where each record links to the previous via cryptographic hash, satisfying 21 CFR Part 11 requirements.

Human Oversight: The requirement that qualified clinical personnel maintain continuous supervisory authority over Physical AI systems during all phases of clinical investigation, with the ability to intervene, override, or terminate operations at any time.

Humanoid Robot: A full-body robotic system with human-like form factor for patient assistance, logistics, and companionship. Evaluated humanoids: Boston Dynamics Atlas (USL 5.8), Agility Digit (USL 4.2), Tesla Optimus (USL 3.6).

Model Context Protocol (MCP): An open protocol for connecting AI agents to external tools and data sources, implemented through five standardized servers (authz, fhir, dicom, ledger, provenance) with 23 tools under Linux Foundation AaIF governance.

MCP Server: One of five domain-specific servers in the national MCP-PAI standard: trialmcp-authz (authorization and RBAC), trialmcp-fhir (FHIR R4 clinical data with de-identification), trialmcp-dicom (DICOM imaging with role-based restrictions), trialmcp-ledger (hash-chained audit), and trialmcp-provenance (DAG-based data lineage).

Physical AI: The integration of artificial intelligence with physical robotic systems and sensor technologies to perform tasks in the real world, encompassing the full pipeline from simulation-based training through real-world clinical deployment.

Pre-Procedure Safety Matrix: The set of eight verification checks (authorization, identity, clinical data, imaging, robot readiness/USL, environment, simulation validation, digital twin sync) that must all pass before a Physical AI system may begin a clinical procedure.

Predetermined Change Control Plan (PCCP): A documented plan for managing modifications to adaptive Physical AI systems, specifying anticipated changes, validation methodology, acceptance criteria, and safety impact assessment per FDA AI/ML guidance.

Reinforcement Learning (RL): A machine learning paradigm for learning through environment interaction and reward signals. In oncology trials, RL trains surgical autonomy, dose optimization, and adaptive protocols via GPU-accelerated training in NVIDIA Isaac Lab (v2.3.1).

Robot Agent: An autonomous Physical AI system executing clinical procedures through the six-step MCP workflow: authenticate, retrieve clinical data, access imaging, execute procedure, record audit, and record provenance.

Robot Capability Profile: A machine-readable JSON specification of a Physical AI platform’s capabilities, safety prerequisites, USL score, and required MCP tools, registered with the authorization server before clinical use.

Secure Aggregation: A cryptographic protocol allowing multiple parties to compute sums of private inputs without revealing individual values, protecting site-specific model updates during federated aggregation.

Sim-to-Real Transfer: The deployment of simulation-trained robotic policies to physical hardware. The sim-to-real gap (performance difference between simulation and reality) must be minimized and documented for clinical applications.

Simulation Framework: A physics-based computational environment for validating Physical AI behavior. Recognized frameworks: NVIDIA Isaac Lab (v2.3.1), MuJoCo (v3.4.0), Gazebo Sim (v10.0.0), PyBullet (v3.2.5).

Surgical Robot: A robotic system for surgical procedures, typically teleoperated. Evaluated surgical robots: da Vinci dVRK (USL 7.1), Hugo RAS (USL 4.5), Versius (USL 3.4).

Task Order: A scheduled clinical trial task assigned to a Physical AI system with defined lifecycle states: scheduled, safety_check, in_progress, completed, cancelled, or failed.

Unification Standard Level (USL): A scoring framework (v1.4.0 through v1.8.0) evaluating robotic platform readiness across four dimensions (simulation switching, AI integration, cross-robot sharing, clinical trial collaboration), with scores from 1.0 (Minimal) to 10.0 (Reference) in three bands: Foundational (1.0 to 4.9), Intermediate (5.0 to 6.9), Advanced (7.0 to 10.0).

Vision-Language-Action (VLA) Model: A multimodal AI model taking visual observations and natural language instructions as input to produce robotic actions, enabling natural language programming of clinical robotic behaviors (NVIDIA GR00T N1.6).

Bibliography

- [1] U.S. Food and Drug Administration. *21 CFR Part 50 – Protection of Human Subjects*. Public domain under 17 U.S.C. § 105. Prior regulation referenced throughout this adaptation. 2024. URL: <https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-50>.